RESEARCH DIARY CONTENT

Weeks 7, 8 and 9 · Capstone Project 1 (Semester VII)

AirAuth — Two-Step Air Authentication
Static Hand Gesture + Dynamic Air Signature

Week	Diary phase	Diary pages
Week 7	Phase 3 · Requirements, Concept Selection & Design	Log p. 43 · sections 3.1–3.4 (pp. 30–33)
Week 8	Phase 3 · Requirements, Concept Selection & Design	Log p. 44 · sections 3.5–3.8 (pp. 34–40)
Week 9	Phase 4 · Implementation & Unit Testing (O1–O2)	Log p. 56 · sections 4.1–4.5 (pp. 45–54)
Evidence	Phase 4 · Technical Evidence (O1–O2)	Requirements p. 47 · Data & Model p. 49 · Security & Quality p. 51

How to use: copy each block into the matching box of the diary by hand. Replace every **[...]** with your real names, student IDs, dates and commit IDs. Change the member tasks and percentages if your team's real work differed. Never fill the **Guide Feedback** box yourselves — the guide writes it during the meeting.

The five objectives (diary pp. 9 & 14) — referenced every week

ID	Objective	Measurable success criterion	Sem
O1	Static hand-gesture capture & check (Step 1)	21 hand points captured in the browser; gesture error rate (EER) ≤ 2% against other people	VII
O2	Air-signature capture & check (Step 2)	Fingertip path captured; signature error rate (EER) ≤ 15% on recorded data (8 registration samples)	VII
O3	AI models + training / calibration pipelines	Two trained neural networks; one-command training; calibrated thresholds saved	VIII
O4	MERN website with two-step login	Register → Step 1 → Step 2 → access granted; lockout, rate limit, login tokens; end-to-end tests pass	VIII
O5	Evaluation, security & real-user testing	FAR / FRR / EER / ROC-AUC reports; replay & step-skip protection; ≥ 10 real users tested	VIII

WEEK 07 — Phase 3 — Requirements, Concept Selection & Design

Weekly Progress Log (diary p. 43)

Week No.	07
Phase	Phase 3 — Requirements, Concept Selection & Design
Dates (from – to)	[dd/mm/yyyy – dd/mm/yyyy]
Planned agenda / focus for the week	Act on the Review 1 feedback. Write the requirements specification (3.1), list the constraints and standards (3.2), come up with candidate solutions (3.3), and score them in a decision matrix (3.4).

Member-wise work done this week

Member (Name)	Student ID	Tasks undertaken this week	Prog. %
1. [Name]	[ID]	Wrote the AI requirements: accuracy targets, response time, number of registration samples. Studied what MediaPipe hand tracking outputs (21 hand points, left/right hand).	100
2. [Name]	[ID]	Wrote the security and data requirements: steps in fixed order, account lockout, login tokens, deleting user data. Listed the standards that apply (ISO 29148, ISO 25010, ISO 27001/OWASP, DPDP Act).	100
3. [Name]	[ID]	Wrote the usability requirements: countdowns, auto-stop, retry messages. Drafted 3 candidate designs and built the scoring matrix with the team.	100

Actionable items completed

- Requirements specification written: 16 requirements, each linked to O1–O5.
- Constraints and standards table completed.
- Five candidate designs listed, and the top three scored in the decision matrix — Concept B selected.

Challenges faced & how they were addressed

- **Problem:** there is no public dataset of people's personal hand gestures to train on.
  Solution: train the gesture AI on computer-generated hand poses, and keep the real recordings for testing only.
- **Problem:** we have real signature recordings from only 3 users.
  Solution: added a requirement for the system to learn from each successful login, and for a tool that exports newly collected data so the models can be retrained.

Evidence produced this week

- Requirements table (diary p. 30).
- Decision matrix (diary p. 33).
- Draft system architecture document: Architecture/01_System_Architecture.md
- Commit ID: [commit id]

Guide feedback / comments

Leave this box blank — your guide writes it. Points to raise in the meeting: • Approval of the five objectives and their targets. • Is training on computer-generated data acceptable if testing uses real recordings? • Confirm the list of standards.

Action plan for next week

Task	Expected outcome	Owner	Due date
Draw the architecture, data-flow and component diagrams	Sections 3.5(a), (b), (c) done	[Member 2]	[date]
Detailed design of O1 and O2	Section 3.6 table done	[Member 1]	[date]
Risk register and work plan	Sections 3.7 and 3.8 done	[Member 3]	[date]

3.1 Requirements specification (diary p. 30)

Req. ID	Type	Requirement	Traces to
FR-01	Functional	Capture the 21 hand points in every frame from a normal webcam, inside the browser	O1
FR-02	Functional	Register a hand gesture from 5 captures of the same pose	O1

Req. ID	Type	Requirement	Traces to
FR-03	Functional	Verify the hand gesture; reject the wrong hand and a different gesture	O1
FR-04	Functional	Record the index fingertip's path (time, x, y), stopping automatically when the finger rests	O2
FR-05	Functional	Register an air signature from 8 samples	O2
FR-06	Functional	Verify the air signature using both its shape and how it is drawn	O2
FR-07	Functional	Login must go Step 1 then Step 2; Step 2 cannot be skipped	O4
FR-08	Functional	Send unregistered usernames to the registration page	O4
FR-09	Functional	Users can view their login history and delete all their biometric data	O4
FR-10	Functional	Retrain and recalibrate the AI models from collected data with one command	O3
NFR-01	Non-functional (performance)	Each verification responds in under 1 second	O1, O2
NFR-02	Non-functional (accuracy)	Gesture error rate (EER) ≤ 2%; signature error rate (EER) ≤ 15%	O1, O2, O5
NFR-03	Non-functional (security)	Lock the account after 5 failures; limit request rates; reject replayed data	O4, O5
NFR-04	Non-functional (privacy)	Video never leaves the browser; stored biometric profiles are never sent to the browser	O4
NFR-05	Non-functional (hardware)	Works with a normal laptop webcam on a regular CPU (no graphics card)	O1, O2
NFR-06	Non-functional (maintainability)	Separate services; automated tests for the AI and the API	O3, O4

3.2 Constraints & standards (diary p. 31)

Constraint / standard	Type	How it shapes the design
Normal webcam with no depth sensor	Hardware	Use MediaPipe's flat (2-D) hand points; treat depth as unreliable
CPU-only machines	Hardware	Keep AI models small (about 100,000 parameters) and optimised for speed
Only 3 real users of signature data	Data	Train on computer-generated data; keep the real data for testing
Two-semester timeline	Schedule	Build the simpler checks first (O1, O2 in CP1), add the neural networks in CP2 (O3)
ISO/IEC/IEEE 29148	Standard	Requirements written with IDs and linked to objectives
ISO/IEC/IEEE 42010	Standard	Architecture described in three views
ISO/IEC 25010	Standard	Quality goals: performance, security, usability, maintainability
ISO/IEC 27001 / OWASP Top 10	Standard	Login tokens, rate limiting, input checking, secrets kept out of the code
ISO/IEC/IEEE 29119	Standard	Test plan from unit to integration to full system, with test IDs
India DPDP Act 2023 / GDPR	Legal	User consent, data used only for login, right to delete biometric data
ACM/IEEE-CS Code of Ethics	Ethics	Report accuracy limits honestly; no real banking data

3.3 Candidate solutions (diary p. 32)

Concept	Description	Pros	Cons
A	Air signature only, as a desktop app (our earlier research prototype)	Simple; already built	Only one factor; not a website; no user accounts
B (selected)	Two steps (hand gesture, then air signature). React website with hand tracking in the browser; Express + MongoDB backend; separate Python AI service	Two factors; video stays on the user's device; any webcam works; built in separate parts	More parts to build and connect
C	Upload video to the server and run a large deep-learning model on the raw frames	Could be very accurate with lots of data	Needs a big dataset and a graphics card; privacy risk; slow

Concept	Description	Pros	Cons
D	Password + air signature	Familiar to users	Keeps passwords, which is the problem we want to remove
E	Gesture login with a depth camera or Leap Motion sensor	Accurate 3-D hand data	Needs special hardware most users don't have

3.4 Decision / trade-off matrix (diary p. 33)

Scores run from 1 (poor) to 5 (excellent). Weighted total = sum of (weight × score).

Criterion	Weight	Concept A	Concept B	Concept C
Security (factors, attack resistance)	0.20	2	5	3
Accuracy with little data	0.15	3	4	1
Privacy	0.15	4	5	1
Hardware cost / accessibility	0.10	4	5	2
Speed	0.10	4	4	2
Feasible in 2 semesters	0.15	5	4	2
Usability / demo value	0.10	2	5	3
Easy to scale and maintain	0.05	2	4	3
Weighted total	**1.00**	**3.35**	**4.60**	**1.95**

Selected concept & justification: Concept B scored highest (4.60). It uses two independent factors: a secret hand pose (something you know) and the way you sign in the air (something you are). Video never leaves the user's device, and it works with any ordinary webcam. It can also be built step by step: the simpler gesture and signature checks (O1, O2) in CP1, and the neural network models (O3) in CP2.

WEEK 08 — Phase 3 — Requirements, Concept Selection & Design

Weekly Progress Log (diary p. 44)

Week No.	08
Phase	Phase 3 — Requirements, Concept Selection & Design
Dates (from – to)	[dd/mm/yyyy – dd/mm/yyyy]
Planned agenda / focus for the week	Design the whole system as three diagrams. Write the detailed design of O1 and O2, the risk register and the work plan. Set up the code repository.

Member-wise work done this week

Member (Name)	Student ID	Tasks undertaken this week	Prog. %
1. [Name]	[ID]	Detailed design of O1: cleaning the hand data, the 115 hand measurements, and the matching method. Detailed design of O2: cleaning the fingertip path, signature matching (DTW) and the habit statistics.	100
2. [Name]	[ID]	Drew the high-level architecture and component diagrams. Designed the MongoDB collections. Set up the repository and folder structure (client/, server/, ml-service/).	100
3. [Name]	[ID]	Drew the data-flow diagram (camera → hand points → server → AI → decision). Wrote the risk register and the work plan for weeks 9–15 and CP2.	100

Actionable items completed

- Three architecture diagrams drawn: high-level, data-flow, component interaction (3.5).
- Detailed design table for O1 and O2 completed (3.6).
- Risk register with 8 risks (3.7) and work plan (3.8) completed.
- Git repository set up with the agreed folder structure.

Challenges faced & how they were addressed

- **Problem:** should the AI run in Node.js or Python?
  Solution: in a separate Python service. The Express backend is the only part users talk to, and it calls the AI service using a secret key.
- **Problem:** hands are different sizes and at different distances from the camera.
  Solution: every hand is converted into its own coordinate system based on the palm, so size, position and tilt no longer matter.

Evidence produced this week

- Architecture diagrams (diary pp. 35–37).
- System architecture document: Architecture/01_System_Architecture.md
- Repository set-up commit: [commit id]

Guide feedback / comments

Leave this box blank — your guide writes it. Points to raise in the meeting: • Ask the guide to review the three diagrams before coding starts. • Agree how many unit tests are expected for the Preliminary Design Review.

Action plan for next week

Task	Expected outcome	Owner	Due date
Write the hand-data cleaning and the 115 hand measurements	Code files preprocessing.py and features.py	[Member 1]	[date]
Write the gesture matcher (register + verify)	Code file verifier.py	[Member 1]	[date]
Hand tracking working in the browser	Live hand skeleton on screen	[Member 3]	[date]
Write the unit test plan	Test IDs UT-01 to UT-15	[Member 2]	[date]

3.5 System architecture (diary p. 34) — short note

The system has three parts. The **website** (React) runs hand tracking in the browser. The **backend** (Express + MongoDB) manages users and the two-step login. The **AI service** (Python) checks the gesture and the signature. Only hand coordinates leave the user's computer, never video.

3.5(a) High-level architecture diagram (diary p. 35) — draw this

```
       [ User + Webcam ]
              |  video (stays in the browser)
              v
+------------ WEBSITE (React, in browser) ------------+
| MediaPipe hand tracking -> 21 hand points per frame |
| Gesture capture screen  .  Air-signature screen     |
+----------------------+-----------------------------+
                       | hand coordinates only (JSON)
                       v
+-------- BACKEND (Express, Node.js) --------+     +-----------+
| Registration . Two-step login              |<--->|  MongoDB  |
| Account lockout . Login tokens . Logs      |     | database  |
+----------------------+---------------------+     +-----------+
                       | secret service key
                       v
+-------- AI SERVICE (Python) --------+
| Step 1: gesture checker             |
| Step 2: air-signature checker       |
| Trained models + thresholds         |
+-------------------------------------+
```

Legend: boxes = separate programs; arrows = data moving between them. Video never leaves the browser.

3.5(b) Data-flow / block diagram (diary p. 36)

```
Camera frame -> MediaPipe -> 21 hand points
   |
   +-- STEP 1 (gesture)
   |     15 frames -> clean + "is the hand still?" check
   |     -> convert to palm coordinates -> 115 hand measurements
   |     -> compare with registered gesture -> score
   |     -> score below threshold?  ->  MATCH / NO MATCH
   |
   +-- STEP 2 (air signature)
         fingertip position + time, every frame
         -> remove still start/end -> remove tracking glitches
         -> resample to 128 points in time -> normalise size and position
         -> DTW shape comparison + 12 habit statistics -> combined score
         -> score below threshold?  ->  ACCESS GRANTED / DENIED
```

3.5(c) Component-interaction diagram (diary p. 37)

Draw four columns — **Website | Backend | AI service | Database** — and these numbered arrows:

#	From → To	Message
1	Website → Backend	"Start login" with the username
2	Backend → Database	Create a login session at Step 1
3	Website → Backend	Gesture sample
4	Backend → AI service	Verify the gesture
5	AI service → Backend	Result (match / no match + score)
6	Backend → Database	Save the attempt; move the session to Step 2
7	Website → Backend	Air-signature sample
8	Backend → AI service	Verify the signature
9	AI service → Backend	Result
10	Backend → Database	Save the attempt, update the profile, close the session
11	Backend → Website	Login token — ACCESS GRANTED

3.6 Detailed design of CP1 objectives O1–O2 (diary p. 38)

Component	Description	Inputs	Outputs	Dependencies
Hand tracker (website)	MediaPipe finds one hand in each video frame	Webcam frames	21 points (x, y, z) + left/right hand	MediaPipe
Gesture cleaning (O1)	Removes bad frames, corrects the camera's width/height ratio, rejects a moving hand, mirrors left hands	15 frames	Clean hand frames	NumPy

Component	Description	Inputs	Outputs	Dependencies
Gesture measurements (O1)	Converts the hand into palm coordinates; measures joint angles, fingertip distances, finger spread, palm direction, bone proportions	Clean frames	115 numbers (middle value over the frames)	Gesture cleaning
Gesture checker (O1)	Checks the 5 registration samples match each other; compares a new gesture with them; rejects wrong hand and replays	Saved profile + new gesture	Accept/reject + score	Gesture measurements
Path cleaning (O2)	Removes still start/end, fixes tracking glitches, resamples to 128 points in time, normalises size and position	Fingertip points with time	128-point path	NumPy
Signature measurements (O2)	9 values per point (position, speed, direction, curvature…) and 12 habit statistics (duration, pauses…)	128-point path	Measurement tables	Path cleaning
Shape matcher, DTW (O2)	Compares paths while allowing for drawing faster or slower in places; uses the closest registered signature	New path + registered paths	Distance	Path cleaning
Signature checker (O2)	Adjusts for each user's own variation, combines scores, applies threshold, rejects replays	Saved profile + new signature	Accept/reject + score	DTW, measurements

3.7 Risk register (diary p. 39)

Risk ID	Description	Likelihood	Impact	Mitigation	Owner
R1	Too little real data to train the neural network models	High	High	Train on computer-generated data; use real data for testing	[Member 1]
R2	Hand tracking jitters or jumps	High	Medium	Automatic glitch correction; middle value over many frames	[Member 1]
R3	Bad lighting or hand outside the camera view	Medium	Medium	"Hand detected" indicator; clear retry messages	[Member 3]
R4	Someone watches and copies the gesture	Medium	High	Step 2 checks how you sign, which is hard to copy	[Member 1]
R5	Attacker re-sends recorded hand data	Low	High	Reject samples identical to stored ones; limit request rates	[Member 2]
R6	Verification too slow on a normal CPU	Medium	Medium	Small models; fast vectorised code; target under 1 second	[Member 1]
R7	Leak of biometric data	Low	High	No video upload; profiles stay on the server; "delete my data" option	[Member 2]
R8	Delays connecting the three parts	Medium	Medium	Agree the data format early; end-to-end test script	[Member 2]

3.8 Project plan / work breakdown (diary p. 40)

Phase / Task	Owner(s)	Start	End	Dependencies
O1: hand-data cleaning + 115 measurements	[Member 1]	Week 9	Week 9	Design (3.6)
O1: gesture checker + unit tests	[Member 1], [Member 2]	Week 9	Week 10	Measurements
Hand tracking in the browser	[Member 3]	Week 9	Week 10	—
O2: path cleaning + DTW	[Member 1]	Week 10	Week 11	Recorded signatures
O2: signature checker + testing on recorded data	[Member 1], [Member 2]	Week 11	Week 12	DTW
Connect O1 + O2 (section 5.1)	All	Week 13	Week 14	O1, O2
Standards checklist + Preliminary Design Review preparation	All	Week 14	Week 15	—
CP2: neural network models (O3), full website (O4), evaluation (O5)	All	CP2 Wk 1	CP2 Wk 13	CP1 work

WEEK 09 — Phase 4 — Implementation & Unit Testing (O1–O2)

Weekly Progress Log (diary p. 56)

Week No.	09
Phase	Phase 4 — Implementation & Unit Testing (O1–O2)
Dates (from – to)	[dd/mm/yyyy – dd/mm/yyyy]
Planned agenda / focus for the week	Start building O1, the hand gesture check: the hand-data cleaning, the 115 hand measurements and the gesture checker. Get hand tracking working live in the browser. Write the unit test plan.

Member-wise work done this week

Member (Name)	Student ID	Tasks undertaken this week	Prog. %
1. [Name]	[ID]	Wrote the hand-data cleaning code (preprocessing.py): reads the frames, corrects the camera's width/height ratio, rejects a moving hand. Wrote the 115 hand measurements (features.py). Started the gesture checker (verifier.py).	70
2. [Name]	[ID]	Wrote the unit test plan (15 tests). Set up the testing tool (pytest). Wrote and ran the first 4 tests.	60
3. [Name]	[ID]	Got hand tracking working in the browser: webcam + MediaPipe drawing the 21-point hand skeleton live (CameraStage.jsx, handLandmarker.js).	60

Actionable items completed

- The 115 hand measurements work correctly: a straight finger measures about 0–15° at its joints, a curled finger about 80–100°.
- The same gesture repeated scores about 0.6 apart; a different gesture about 3.0 apart — clearly separable.
- The live hand skeleton shows in the browser.
- 4 unit tests written and passing (UT-01, UT-02, UT-08, UT-09).

Challenges faced & how they were addressed

- **Problem:** our test data kept failing the "is the hand still?" check.
  Solution: the test-data generator was moving the hand to a new spot in every frame. We fixed the generator to keep the hand in one place during a capture. The check itself was correct.
- **Problem:** the same gesture gave different numbers when the hand was tilted.
  Solution: every hand is now turned into its own palm-based coordinates before measuring, which removed the effect of tilt.

Evidence produced this week

- Code files: ml-service/airauth/static_gesture/preprocessing.py, features.py, verifier.py (started)
- Browser code: client/src/components/CameraStage.jsx, client/src/lib/handLandmarker.js
- Tests UT-01, UT-02, UT-08, UT-09 (ml-service/tests/test_static_gesture.py)
- Commit IDs: [commit ids]

Guide feedback / comments

Leave this box blank — your guide writes it. Points to raise in the meeting: • Show the live hand skeleton and the measurement values. • Confirm the O1 accuracy target (error rate ≤ 2%) is acceptable.

Action plan for next week

Task	Expected outcome	Owner	Due date
Finish the gesture checker (register, verify, wrong hand, replay)	All O1 tests pass	[Member 1]	[date]
Start the signature path cleaning (O2)	Code file dynamic_signature/preprocessing.py	[Member 1]	[date]
Gesture capture screen with countdown	Code file StaticGestureCapture.jsx	[Member 3]	[date]
Run and record the remaining O1 tests	Results in table 4.4	[Member 2]	[date]

4.1 Implementation progress (diary p. 45)

Task	Owner(s)	Start	End	% done	Dependencies
Hand-data cleaning	[Member 1]	Week 9	Week 9	100	—
115 hand measurements	[Member 1]	Week 9	Week 9	100	Hand-data cleaning
Gesture checker	[Member 1]	Week 9	Week 10	50	Measurements
Hand tracking in the browser	[Member 3]	Week 9	Week 10	60	MediaPipe
O1 unit tests	[Member 2]	Week 9	Week 10	40	Gesture checker
Signature path cleaning (O2)	[Member 1]	Week 10	Week 11	0	—

Update the % column each week as work progresses.

4.2 Deviations from design (diary p. 46)

If nothing changed from the Week 8 design, write **"No deviations this week."** If you want an entry, this one is accurate:

Objective / module	Deviation from design	Reason	Impact
O1 / test data	Added a computer-generated hand model to create test gestures	No public dataset of personal gestures exists	Lets us test O1 without real users; real users are still needed later (O5)

Build & Version Control (diary p. 50) — start this table now

Module / component	Owner	Repository / commit ref	Build status
Gesture cleaning + measurements (O1)	[Member 1]	[commit id]	Tests passing
Browser hand tracking	[Member 3]	[commit id]	Builds and runs in browser
Unit test set-up (pytest)	[Member 2]	[commit id]	4 / 4 passing

4.3 Unit test plan (diary p. 52)

The test IDs line up with the real tests already in ml-service/tests/.

Test-case ID	Description	Objective / module	Test type
UT-01	Measurements have the right size (115 numbers; 89 for the AI)	O1 / measurements	Unit
UT-02	Moving or resizing the hand doesn't change the measurements	O1 / measurements	Unit
UT-03	The correct gesture is accepted (at least 4 of 5 tries)	O1 / gesture checker	Unit
UT-04	A different gesture is rejected	O1 / gesture checker	Unit
UT-05	The wrong hand is rejected	O1 / gesture checker	Unit
UT-06	An exact copy of a stored sample is rejected	O1 / gesture checker	Security
UT-07	Registration finds the one sample that is a different gesture	O1 / gesture checker	Unit
UT-08	A moving hand is rejected	O1 / hand-data cleaning	Unit
UT-09	Too few frames are rejected	O1 / hand-data cleaning	Unit
UT-10	Signature path becomes 128 points and is normalised	O2 / path cleaning	Unit
UT-11	A tracking glitch is found and fixed	O2 / path cleaning	Unit
UT-12	DTW gives 0 for identical paths and the same result both ways	O2 / DTW	Unit
UT-13	The owner's signature is accepted; other people's are rejected	O2 / signature checker	Unit
UT-14	A replayed signature is rejected	O2 / signature checker	Security
UT-15	A too-short signature is rejected	O2 / path cleaning	Unit

4.4 Unit test execution (diary p. 53) — Week 9 entries only

Record only the tests you actually ran this week:

Test-case ID	Input	Expected output	Actual output	Pass/Fail
UT-01	One gesture capture (10 frames)	115 numbers; 89 for the AI	115; 89	Pass
UT-02	Same hand frame, enlarged 1.7× and shifted	Same measurements	Same (difference < 0.000001)	Pass

Test-case ID	Input	Expected output	Actual output	Pass/Fail
UT-08	8 frames with the hand in a new place each frame	Error: "hand moved"	Error: "hand moved"	Pass
UT-09	No frames	Error: "too few frames"	Error: "too few frames"	Pass

Add UT-03 to UT-15 in weeks 10–12 as you run them.

4.5 Issue tracker (diary p. 54)

Issue no.	Description	Impact	Owner	Status	Guide sign
I-01	Test data failed the "hand still?" check because the generator moved the hand every frame	Medium (blocked testing)	[Member 1]	Closed: generator fixed	(guide)
I-02	The same gesture gave different numbers when the hand was tilted	High (wrong matches)	[Member 1]	Closed: palm-based coordinates added	(guide)

Technical Evidence — CSE & Allied (CP1 · objectives O1–O2)

These pages are filled during Phase 4 (weeks 9–12) and checked at the Preliminary Design Review. Every number below comes from the project's real test runs (ml-service/reports/ and the test suite). If your guide wants CP1 pages limited to O1–O2 only, keep the rows marked O1/O2 and move the O4 rows to the CP2 pages (pp. 79–83).

Requirements & Architecture — ISO/IEC/IEEE 29148 · 42010 (diary p. 47)

Req. ID	Type (F/NF)	Requirement / quality attribute	Priority	Traces to
FR-01	F	Capture the 21 hand points in every frame from a normal webcam, inside the browser	High	O1
FR-02	F	Register a hand gesture from 5 captures of the same pose; reject a set where one capture is a different gesture	High	O1
FR-03	F	Verify the hand gesture; reject the wrong hand, a different gesture and an exact replay	High	O1
FR-04	F	Record the index fingertip's path (time, x, y) and stop automatically when the finger rests	High	O2
FR-05	F	Register an air signature from 8 samples; reject a badly captured sample	High	O2
FR-06	F	Verify the air signature using shape (DTW) and drawing habits (speed, pauses, duration)	High	O2
FR-07	F	Give a clear retry message when a capture is unusable (too short, hand moved, too few frames)	Medium	O1, O2
NFR-01	NF · Performance	Each verification responds in under 1 second on a normal CPU	High	O1, O2
NFR-02	NF · Accuracy	Gesture error rate (EER) ≤ 2%; signature error rate (EER) ≤ 15%	High	O1, O2
NFR-03	NF · Security	Reject replayed (copied) samples	High	O1, O2
NFR-04	NF · Privacy	Only hand coordinates are processed — never video	High	O1, O2
NFR-05	NF · Portability	Works with an ordinary laptop webcam, no special hardware	Medium	O1, O2
NFR-06	NF · Maintainability	Each module has automated unit tests	Medium	O1, O2
NFR-07	NF · Robustness	Works regardless of hand size, position and tilt; corrects tracking glitches	High	O1, O2

Architecture view for this page: reuse the high-level diagram from section 3.5(a) (diary p. 48 — quality attributes covered: performance, security, usability, maintainability, portability).

Data & Model Evidence (diary p. 49)

Dataset(s)

Dataset / source	Size & split	Licence & provenance	Pre-processing / issues
Research air-signature dataset (recorded by the team with a desktop webcam app, 1280×720)	3 users × 3 sessions × 10 = 90 signatures. Split by session: register on session 1 (8 signatures), test on sessions 2–3.	Own data, collected from consenting volunteers [confirm names/consent forms]. Not shared publicly.	Fingertip path only. Glitch removal, idle trimming, resampling to 128 points. Issue: only 3 users, and they drew simple shapes.
Computer-generated hand poses (our own kinematic hand model)	2,048 pose types: 1,638 for training, 410 held out for testing. 150 simulated users for evaluation.	Generated by our code (synthetic_hand.py) — no licence restrictions.	Random hand sizes, angles (±35°) and tracking noise. Issue: synthetic, so real-world error may be higher.
MediaPipe Hand Landmarker (pre-trained model by Google)	Used as-is; not retrained.	Apache 2.0 licence (MediaPipe).	Gives 21 hand points per frame. Issue: jitter and occasional jumps, handled in cleaning.

Experiments

Run ID	Config / model	Metric	Baseline	Result vs target
EXP-01	Signature DTW using x,y only vs x,y + depth (z)	EER	x,y,z: 30.8%	x,y only: 14.4% — depth removed from the design

Run ID	Config / model	Metric	Baseline	Result vs target
EXP-02	Signature DTW: compare to closest sample vs average of 3 closest vs average of all (5 registration samples)	EER	Average of all: 27–49%	Closest sample: 18–22% — chosen
EXP-03	Number of registration signatures: 5 vs 10 vs 20 (two sessions)	EER	5 samples: 18–22%	10: 8–12%; 20: 3–6% — registration set to 8, plus learning from logins
EXP-04	Gesture template matcher, 150 simulated users, unseen gestures	EER	Target ≤ 2%	1.2% — target met
EXP-05	Signature check: DTW + habit statistics, 8 registration samples, later sessions	EER	DTW alone: 13.3%	8.7% — target ≤ 15% met
EXP-06	Verification speed (one request, 8 / 20 stored signatures)	Latency	Target < 1 s	≈ 140 ms / ≈ 470 ms — target met
EXP-07	Gesture check vs someone who copies the exact gesture	EER	—	37.7% — expected limitation; Step 2 covers it

Bias / fairness & data-privacy check (consent, anonymisation, DPDP/GDPR relevance)

- **Small, narrow sample:** only 3 real users, so results may not hold for everyone (different hand sizes, skin tones, ages, or left-handed users). Plan: test with 10–20 varied volunteers in CP2 (O5).
- **Hand tracking fairness:** MediaPipe's accuracy can vary with lighting and skin tone. We added a "hand detected" indicator and clear retry messages; to be measured across users in CP2.
- **Left-handed users:** left hands are mirrored into the same layout, so both hands are supported.
- **Consent:** volunteers agree before their signatures are recorded [attach consent forms].
- **Data minimisation:** only hand coordinates are stored — no video, no face images.
- **Anonymisation:** exported training data uses random user IDs, not names; raw data is excluded from the Git repository.
- **DPDP Act 2023 / GDPR:** data is used only for login; users can delete their account and all biometric data.

Security & Quality — OWASP / ISO 27001 · ISO/IEC 25010 (diary p. 51)

Control / quality attribute	Applicable? (Y/N)	How addressed / evidence
Replay attack protection (O1, O2)	Y	A sample identical to a stored one is rejected. Tests UT-06 and UT-14.
Input validation — OWASP A03 Injection (O1, O2)	Y	Every request is checked for shape and size before processing (schemas.py); malformed data returns an error, never crashes.
Privacy of biometric data — ISO 27001 / DPDP (O1, O2)	Y	No video leaves the browser; stored profiles never sent back to the browser; "delete my data" option.
Service access control — OWASP A01 (O1, O2)	Y	The AI service only answers requests carrying the secret service key (security.py); test test_biometric_endpoints_require_service_key.
Brute-force protection — OWASP A07 (O4)	Y	3 attempts per step, account locked after 5 failures, request rate limits.
Step order cannot be skipped — OWASP A01 (O4)	Y	Server tracks which step each login is on; Step 2 rejected unless Step 1 passed (end-to-end test).
Secrets & configuration — OWASP A05	Y	Secrets kept in .env (not in Git); security headers (helmet); CORS limited to the website.
Vulnerable components — OWASP A06	Y	Library versions pinned; run npm audit and pip check before PDR [to do].
Performance efficiency — ISO 25010	Y	Verification ≈ 140–470 ms (EXP-06); target under 1 s met.
Reliability — ISO 25010	Y	Unusable captures give a retry message; if the AI model is missing, the simpler checks still work.
Usability — ISO 25010	Y	Countdowns, live hand skeleton, "hand detected" indicator, auto-stop, clear messages.
Maintainability — ISO 25010	Y	Separate modules for cleaning, measurements and checking; 23 automated tests passing.

Before copying: replace every [...] with real names, student IDs, dates and commit IDs, and change any task or percentage that doesn't match what your team actually did.